Day 1 Problem Set

Methods Camp | University of Texas at Austin

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Day 1 Problem Set

This problem set contains exercises from Day 1 that were originally done in small groups. To reinforce your understanding, please complete these exercises independently.

Exercise 1: Functions and Operations

1.1 Let $f(x) = 5x - 4$. Compute $f(2)$ and $f(0)$.

1.2 Let $f(x, y) = 3x + y^2$. Compute $f(2, 3)$.

1.3 Simplify: $\ln(50) - \ln(10)$. (Hint: use a log rule.)

1.4 Solve for x: $\ln(x) = 2$. (Hint: rewrite in exponential form.)

1.5 A class survey of 6 students records number of siblings: $\{1, 0, 2, 3, 1, 2\}$.

a. Write this as a summation expression and compute $\sum_{i=1}^{6} x_i$.

b. Compute the mean $\bar{x}$.

Exercise 2: Ratios, Proportions, Rates & Percentages

A town has 5,400 women and 4,600 men.

2.1 Compute the sex ratio (males per 100 females).

2.2 Compute the proportion of the town that is female.

2.3 Last year the town recorded 27 burglaries. Compute the burglary rate per 100,000 residents. (Total population = 10,000.)

2.4 The town's unemployment rate fell from 8% to 6%.

a. What is the change in percentage points?

b. What is the percent (relative) change?

2.5 A local news headline says: "Unemployment drops 2%!" Is this headline accurate? Why or why not?

Exercise 3: Logarithms and Income

A researcher studies how income relates to years of education (X), using a model of log income:

$$\ln(Y) = 9.00 + 0.08X$$

where Y is annual income in dollars.

3.1 For a respondent with $X = 12$ years of education, compute $\ln(Y)$.

3.2 Convert $\ln(Y)$ into Y (dollars) by exponentiating. (Hint: use the inverse of ln.)

3.3 Now consider a respondent with $X = 13$. Compute their $\ln(Y)$ and their Y.

3.4 Exponentiate the coefficient on X: $e^{0.08} \approx 1.083$. Interpret this number in one sentence.